EE23B085

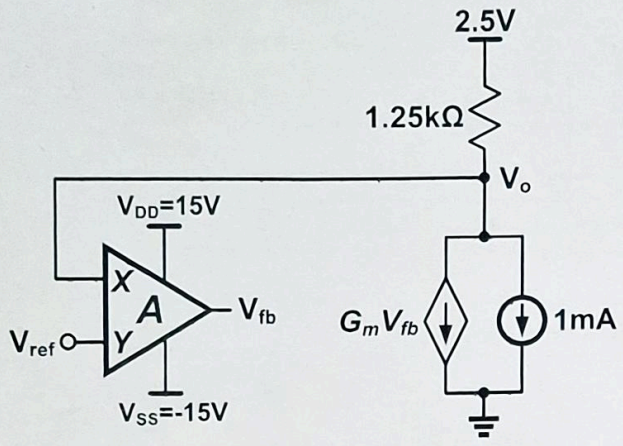
V'Eswar Gupta

Quiz-I (Jan.-May 2025)
EE2019 – Analog Systems and Lab

Total Marks: 15

Duration: 50 min

Question-1 **(5 marks)**

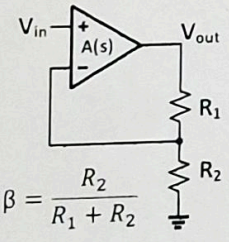


The circuit shown in the figure above operates in negative feedback configuration. Assume opamp gain A=80 and Gm = 10mS, and V_{ref} = 1.25V. The opamp has infinite input impedance and zero output impedance.

a) Determine the signs of the opamp inputs (X,Y) for the opamp operating in negative feedback. State your reasoning carefully. *(2 marks)*
b) Find dc output voltage V_o for V_{ref} = 1.25V. (1 mark)
c) Determine the ac gain V_o/V_{ref}. *(2 marks)*

Question-2 **(5 marks)**

In the non-inverting amplifier shown below, $\omega_{p1} = 10^3$ rad/s, $\omega_{p2} = 10^8$ rad/s, $R_1 = R_2 = 1k\Omega$.



$$\beta = \frac{R_2}{R_1 + R_2}$$

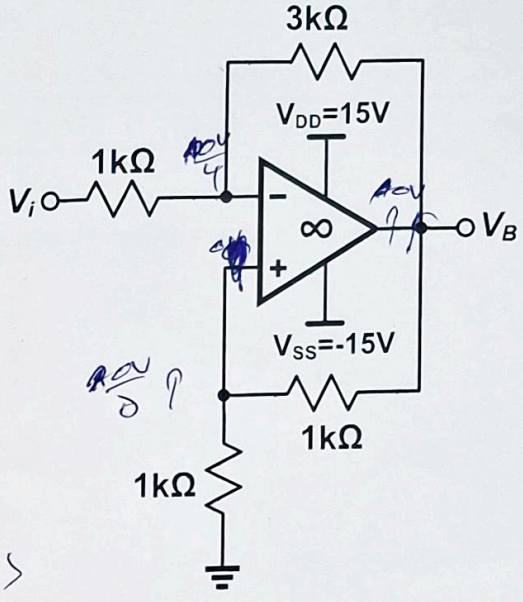
$$A(s) = \frac{A_o}{\left(1 + \frac{s}{\omega_{p1}}\right)\left(1 + \frac{s}{\omega_{p2}}\right)}$$

$$A_o = 10^4$$

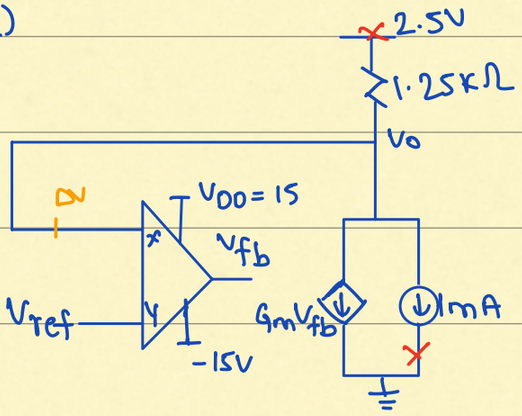
a) Find the open loop gain LG(s) for the above closed loop operation. *(2 marks)*
b) Find the unity gain frequency and phase margin for the above negative feedback loop. *(3 marks)*

Question-3 **(5 marks)**

For the circuit shown below, plot the dc transfer characteristic of V_B vs. V_i, when V_i varies from -15V to +15V. Mark all the values of V_i and V_B properly.



① a)



$A = 80 \quad V_{ref} = 1.25V \quad G_m = 10^{-2} S$

$V_{fb} = 80 (V_o - V_{ref})$

$(V_o)_{final} = (V_o)_{original} - (G_m A \Delta V) . R$

If $x \to +ve$, $Y \to -ve \Rightarrow \Delta V \to$ decreased $\Rightarrow x \to +ve$, $Y \to -ve$

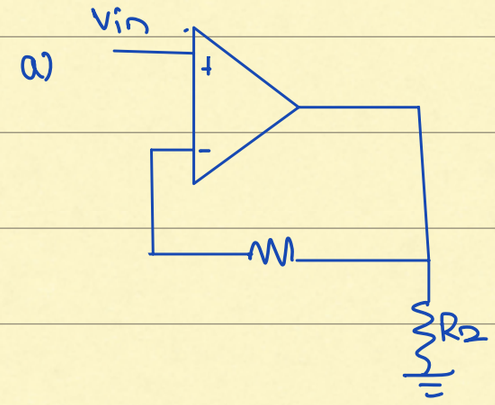
b) 1.25V

c) $-\frac{V_o}{1.25} = 10^{-2} V_{fb} \Rightarrow V_o = -12.5 V_{fb} = -12.5 (80)(V_o - V_{ref})$

$1001 V_o = 1000 V_{ref}$

$\Rightarrow V_o / V_{ref} = 1000/1001$

② a)



$LG = A(s) . \beta$

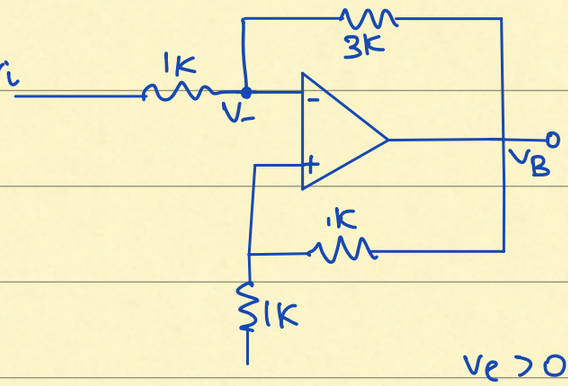
$= \frac{10^4}{(1+\frac{s}{10^3})(1+\frac{s}{10^8})} . \frac{1}{2}$

b) $|LG(j\omega_u)| = 1 \Rightarrow 1 = \frac{5 \times 10^3}{\omega_u^2 / 10^3} \Rightarrow \omega = 5 \times 10^6$ rad/s

$\angle LG(j\omega) = -\tan^{-1}(\frac{\omega}{\omega_{p1}}) - \tan^{-1}(\frac{\omega}{\omega_{p2}}) = -92.9°$

Phase margin $= 180° - |\angle LG(j\omega)| = 87.1°$

③



$$V_- = \frac{V_B}{4} + \frac{3V_i}{4}$$

$$V_+ = \frac{V_B}{2}$$

$$V_e = \frac{V_B}{4} - \frac{3V_i}{4}$$

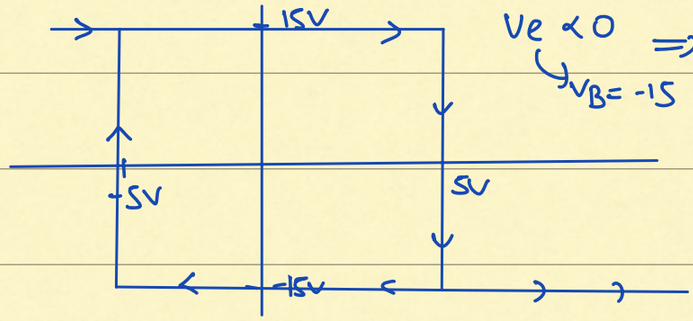
$V_e > 0 \Rightarrow \frac{V_B}{4} > \frac{3V_i}{4}$

$V_B = 15 \Rightarrow V_i < 5$

$V_e < 0 \Rightarrow \frac{V_B}{4} < \frac{3V_i}{4}$

$V_B = -15$

$\Rightarrow V_i > -5$



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B

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Quiz I [✓] Quiz II/ Mid-Sem [ ] End-Semester [ ] Make-up [ ] Date : 20/2/2025

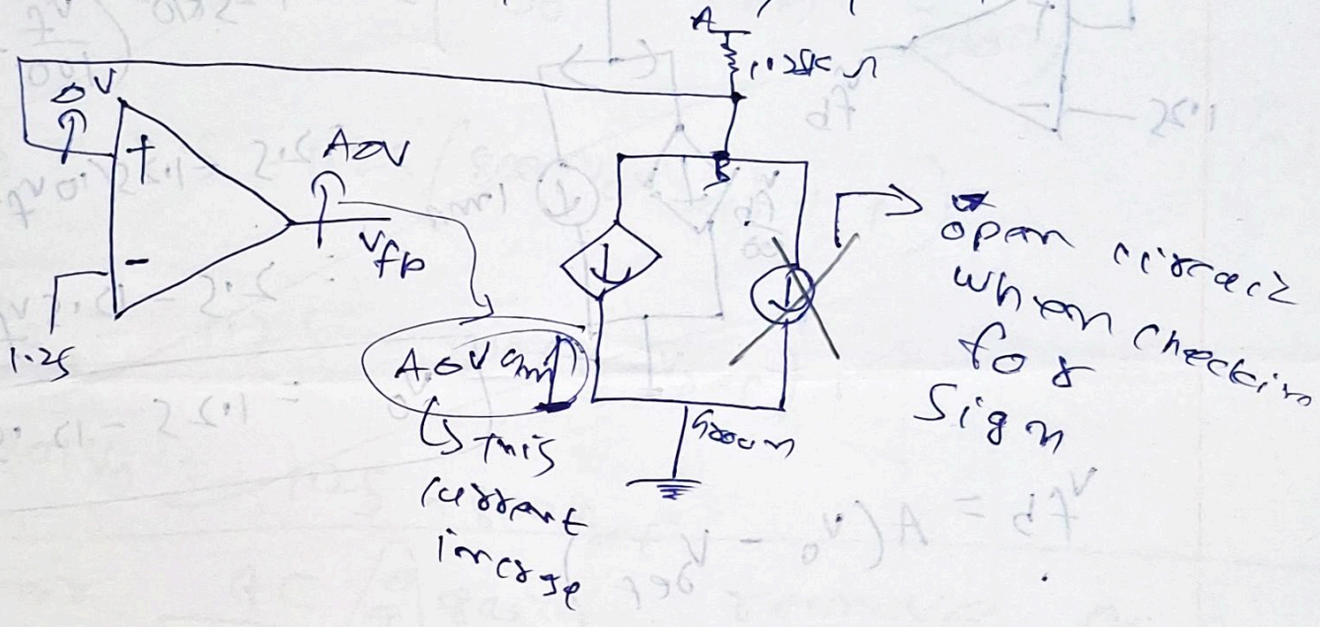
Semester & Degree : 4th - (B-Tech) Course No. EE2019 Part :

Question No.	1	2	3	4	5	6	7	8	9	10
Marks	5	5	5							

11	12	13	14	15	16	17	18	19	20	Total
										15/15

Answer on both sides of the paper including the space below

Let me assume X = +ve, Y = -ve



Due to increase in current in (B-Ground)

$V_B = (V_B)_{original} - (A o V_{gm}) R$

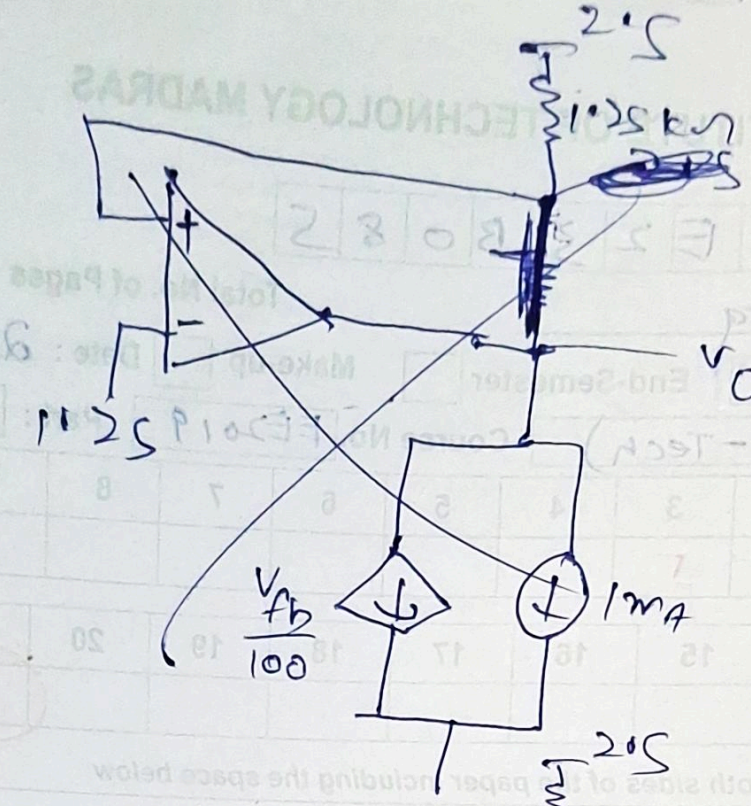
This much voltage decreased at 'B' (ie) at input

given ΔV, got $-A\Delta V_{gm}R$

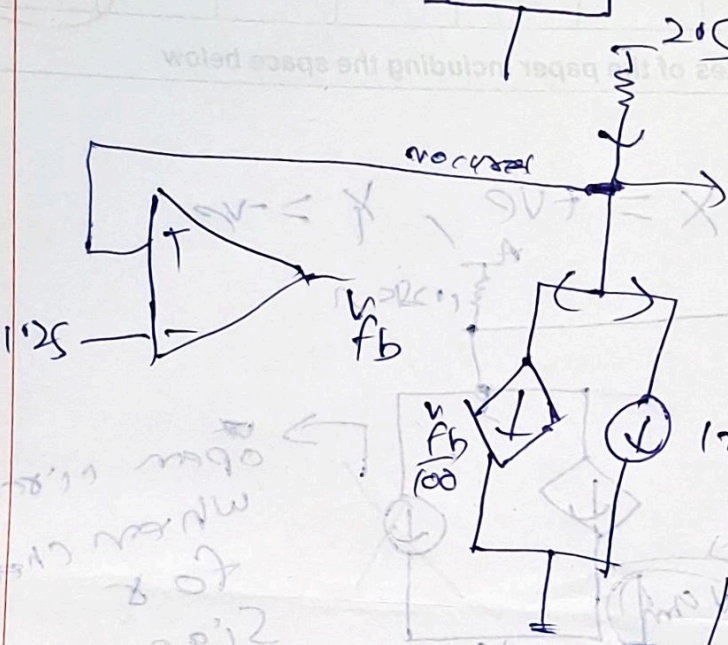
∴ X = +, Y = - is the correct sign for -ve feedback

(2)

(b)



(b)



$$2.5 - 1.25\times10^{3}\left(\frac{V_{fb}}{100} + \frac{1}{100}\right)$$

$$= 2.5 - 12.5(1 + V_{fb}) - 2.5$$

$$= -V_{fb} - 12.5 - 2.5 =$$

$$V_o = -V_{fb} - 15$$

$$V_{fb} = A(V_o - V_{ref})$$

$$V_{fb} = 80(-12.5 - V_{fb})$$

↳ true only if $V_{fb} = 0$

(ii) $V_o = 1.25 - 12.5(0)$

$V_o = 1.25\ V$

(1)

(c)

$$V_o = 2.5 - 1.25\times10^{3}\left(G_m V_{fb} + 1mA\right)$$

$$V_{fb} = A\left(V_o - V_{ref}\right)$$

$$V_o = 2.5 - 1.25\times10^{3}\left(G_m A(V_o - V_{ref}) + 10^{-3}\right)$$

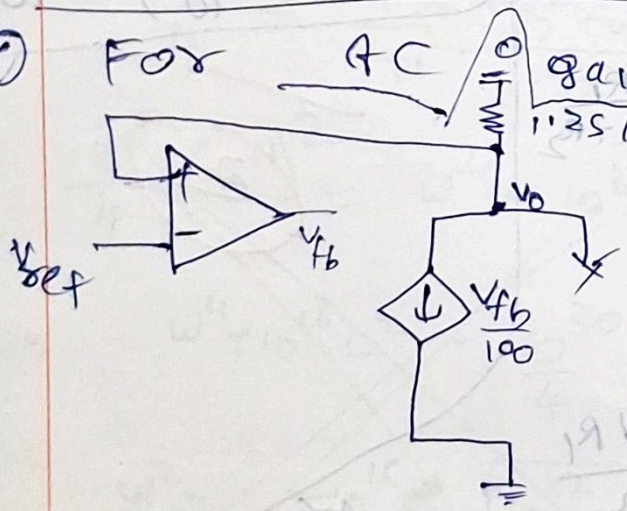
$$V_o = 2.5 - 1.25\times10^{3}\left(\frac{80}{100}(V_o - V_{ref}) + \frac{1}{1000}\right)$$

$$V_o = 2.5 - 1.25\left(800(V_o - V_{ref}) + 1\right)$$

$$V_o = 1.25 - 800(V_o - V_{ref})$$

$$801 V_o = 1.25 + 800 V_{ref}$$

(d) For AC gain remove DC-source



$$V_o = -\frac{V_{fb}}{100}(1.25\times10^{3})$$

$$\boxed{V_o = -12.5\,V_{fb}}$$

$$V_{fb} = A(V_o - V_{ref}) = 80(-12.5 V_{fb} - V_{ref})$$

$$V_{fb} = -1000\,V_{fb} - 80 V_{ref}$$

$$1001\,V_{fb} = -80 V_{ref}$$

$1001 V_{fb} = -80 V_{ref}$ $\quad V_o = -12.5 V_{fb}$

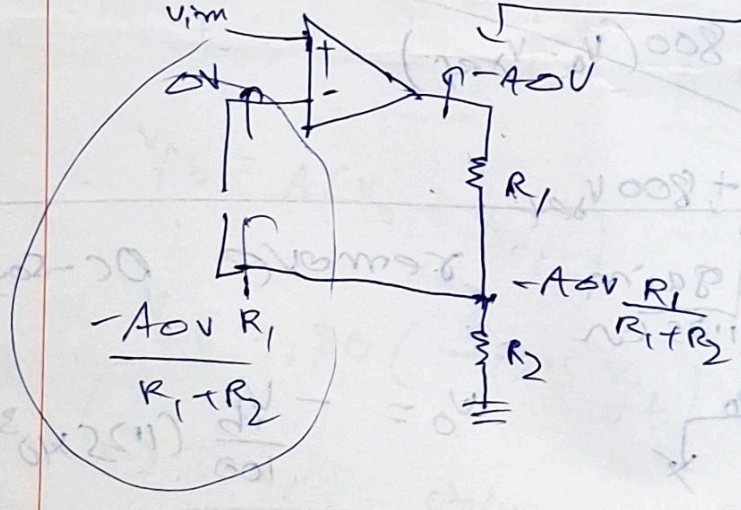
$1001\left(\frac{V_o}{-12.5}\right) = -80\, V_{ref}$

$$\frac{V_o}{V_{ref}} = \frac{(80)(12.5)}{1001} = \frac{1000}{1001}$$

∴ gain is $\frac{1000}{1001}$ (2)

(2)

a) $L_G(s) = A(s)\,\beta$ ⇒ Proof



$$L_G(s) = \frac{10^4\left(\frac{1}{2}\right)}{\left(1+\frac{s}{10^3}\right)\left(1+\frac{s}{10^8}\right)}$$

(2)

$-L_G(s) = \dfrac{-A0V\,R_1}{R_1+R_2}$ $\quad = -A$

as it is in negitive feedback

$\beta = \frac{1}{2}$

(2) $L_G(s) = A(s)\,\beta$

(b) as $\omega_{p1} = 10^3$, $\omega_{p2} = 10^8$ Dominate Pole effect is in bray

$\therefore \omega_{GB} = A_0 \beta P_1$

$= 10^4 \left(\frac{1}{2}\right)(10^3)$

$= \frac{10^7}{2} = 5 \times 10^6$ rad/sec (1.5)

without dominant pole approximation (even then you get almost same ans)

$$|A(s)\beta| = \frac{10^4}{2} \frac{}{\sqrt{\frac{\omega^2}{\omega_{p1}^2}+1}\sqrt{1+\frac{\omega^2}{\omega_{p2}^2}}} = 1$$

$$\frac{10^8}{4} = \frac{(\omega^2+\omega_{p1}^2)(\omega_{p2}^2+\omega^2)}{\omega_{p1}^2 \omega_{p2}^2}$$

$$\frac{10^8}{4}(10^{11})^2 = \omega^4 + 10^6\omega^2$$

$$\omega^4 + 10^{16}\omega^2 - \frac{10^{30}}{4} = 0$$

$$\omega^2 = \frac{-10^{16} \pm \sqrt{10^{32}+10^{30}}}{2}$$

$$\frac{10^{30}}{4} = \omega^4 + \omega^2(10^{16}+10^6) + 10^{22}$$

$$\omega^4 + \omega^2(10^{16}+10^6) + 10^{22} - \frac{10^{30}}{4}$$

$$\frac{-(10^{16}+10^6) + \sqrt{(10^{16}+10^6)^2 - 4(10^{22} - \frac{10^{30}}{4})}}{2}$$

$$\frac{-(10^{16}+10^6) + \sqrt{(10^{16}+10^6)^2 - 4(10^{22} - \frac{10^{30}}{4})}}{2}$$

$\omega^2 = 2.4993 \times 10^{13}$

$\omega = 4.9993 \times 10^6$ rad/sec

Phase margine

$$= 180 - \tan^{-1}\left(\frac{\omega_{ugf}}{P_1}\right) - \tan^{-1}\left(\frac{\omega_{ugf}}{P_2}\right)$$

$$= 180 - 90 - \tan^{-1}\left(\frac{5\times10^6}{108}\right)$$

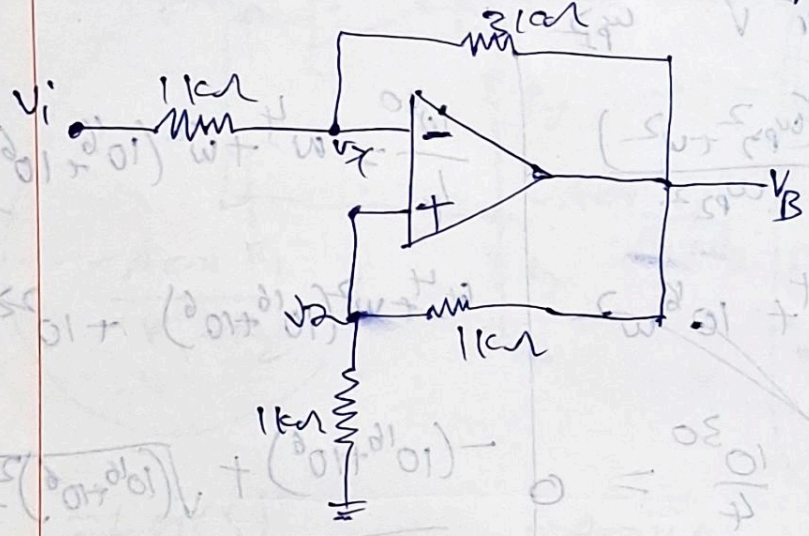
$$= 180 - 90 - 2.858$$

$$= 87.1411$$

Phase margin = 87.1411° (1.5)

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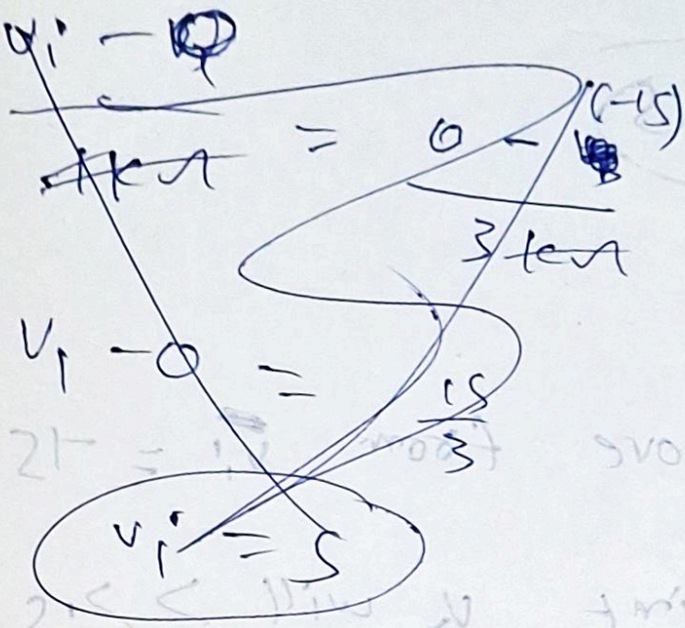
(3) The above circuit is in positive feedback



If V_i is large positive value then V_B = + ve saturation V_x is a large positive value only If V_2 is large +ve then V_{id} = large negative $\therefore V_B$ = -ve hence for large positive V_B = -ve

moving from +15 toword -15

Finding at what v_i the v_x will be



$$\frac{v_i - \left(\frac{-15}{2}\right)}{} = \frac{\frac{-15}{2} - (-15)}{}$$

$$v_i + \frac{15}{2} = \frac{5}{2}$$

$$\boxed{v_i = -5}$$

If $v_i < -5$ then if $\boxed{V_B = +15}$

then $v_x < -7.5$

$\therefore \quad v_a - v_x > 0 \qquad \boxed{V_B = +15}$

it will flip

If v_i is large -ve value

v_x = large negitive

v_{id} = large positive

$\therefore$ $V_B = +1S$

$v_2 = +7.5$

as you move from $v_i = -1S$ to $1S$

at some point v_2 will > 7.5 which will cause v_{id} = -ve $v_{out} = -1S$

$$\frac{v_i' - (7.5)}{1K} = \frac{7.5 - 1S}{3K}$$

$$3v_i' - 22.5 = -7.5$$

$$3v_i' = 1S \Longrightarrow \boxed{v_i' = 5}$$

→ flipping point

